

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (Automobile)

as per NEP 2020

w.e.f session 2026-27

Course Outline for B.Voc (Automobile)
as per NEP 2020

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 101	Core Discipline	Elements of Automobile Technology-I	2	2	4	40	10	40	10	100
AM 102	Core Discipline	Automotive Electronics and Electric System	2	2	4	40	10	40	10	100
AM 103	Core Discipline	Automotive Tools	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC I	Skill Enhancement Course	Common Basket			3					75
IDC I	Interdisciplinary Course	Common Basket			3					75
VAC I	Value-Added Course	Common Basket			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 201	Core Discipline	Elements of Automobile Technology-II	2	2	4	40	10	40	10	100
AM 202	Core Discipline	Advanced Electronic and Auto Electric System	2	2	4	40	10	40	10	100
AM 203	Core Discipline	Automobile Physics	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC II	Skill Enhancement Course	Common Basket			3					75
IDC II	Interdisciplinary Course	Common Basket			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - **Assistant Auto Technician**

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Automobile = 48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 301	Core Discipline	Thermodynamical Physics	4	-	4	80	20	-	-	100
AM 302	Core Discipline	Automobile Technology-I	2	2	4	40	10	40	10	100
AM 303	Core Discipline	Fundamentals of Petrol Engine	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC III	Skill Enhancement Course	Common Basket			3					75
IDC III	Interdisciplinary Course	Common Basket			3					75
VAC II	Value-Added Course	Common Basket			2					50
PW II	Project Work	Project Work-II & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 401	Core Discipline	Lubrication System, Supercharging and Electric Vehicles	4	-	4	80	20	-	-	100
AM 402	Core Discipline	Automobile Technology-II	4	-	4	80	20	-	-	100
AM 403	Core Discipline	Fundamentals of Diesel Engine	2	2	4	40	10	40	10	100
AM 404	Core Discipline	Workshop -I	-	4	4	-	-	80	20	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
VAC III	Value-Added Course	Common Basket			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - **Automotive Service Technician**

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits Required to get Diploma Course in Automobile = 96 + 4 #.

#Exit 2: Diploma Course - Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 501	Core Discipline	Solar Cells in Automobile Technology	4	-	4	80	20	-	-	100
AM 502	Core Discipline	Automobile Technology-III	4	-	4	80	20	-	-	100
AM 503	Core Discipline	Workshop 2	-	4	4	-	-	80	20	100
AM 504^	Core Discipline	Modern Electric and Hybrid Vehicles-I	2	2	4*	40	10	40	10	100
AM 505^	Core Discipline	Off Road Vehicles and Automotive Fuels	2	2	4*	40	10	40	10	100
AM 506^	Core Discipline	Advanced Automotive Fault Diagnosis-I	2	2	4*	40	10	40	10	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work-III & Viva-Voce			2					50
Total Credits					**24**					**600**

^Any two in 5th Semester

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 601	Core Discipline	Modern Vehicle Technology	4	-	4	80	20	-	-	100
AM 602	Core Discipline	Automobile Technology-IV	4	-	4	80	20	-	-	100
AM 603	Core Discipline	Vehicle Performance and Garage Setup	2	2	4	40	10	40	10	100
AM 604	Core Discipline	Workshop 3	-	4	4	-	-	80	20	100
AM 605^	Core Discipline	Modern Electric and Hybrid Vehicles II	2	2	4*	40	10	40	10	100*
AM 606^	Core Discipline	Automobile Electrical System	2	2	4*	40	10	40	10	100*
AM 607^	Core Discipline	Advanced Automotive Fault Diagnosis - II	2	2	4*	40	10	40	10	100*
Total Credits					**24**					**600**

^Any two in 6th Semester

Job Role - **Assistant Service Supervisor**

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (Automobile) (SEMESTERS I & II)

PROGRAMME OUTCOMES

PO1. Understanding of the basic concepts and principles in automobiles.

PO2. Knowledge about automobile component both mechanical and electrical/electronic.

PO3. Understanding and practical experience of the automotive tools and equipment's.

PO4. Understanding of concepts and principles in automotive electronic system.

PO5. Adequate Professional Skills and Competencies in measuring instruments.

PO6. Knowledge about operation/working of engines along with its components.

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (Automobile) (SEMESTERS III & IV)

PROGRAMME OUTCOMES

PO1. Understanding of concepts and laws in automotive thermodynamic system.

PO2. Understanding of the basic concepts and principles in Clutch and Gearbox.

PO3. Understanding and practical experience of the automobile steering system and drive line.

PO4. In depth knowledge about Petrol Engines and Diesel Engines.

PO5. Adequate Professional Skills and Competencies Lubrication system and Cooling system of automobiles.

PO6. Knowledge about operation/working of Electric Vehicles.

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (Automobile) (SEMESTERS V & VI)

PROGRAMME OUTCOMES

PO1. Understanding the basic concepts and principles of Modern Technologies in Automobiles.

PO2. Knowledge about Braking, Suspension and Air Conditioning in detail.

PO3. Understanding and practical experience of automotive denting and painting.

PO4. In depth knowledge about Electric and Hybrid Vehicles.

PO5. Adequate Professional Skills and Competencies in parameters related to vehicle performance.

PO6. Knowledge about operation/working of electrical components and advance diagnosing techniques.

Semester I

<table>
<tr><th>1</th><th>Course Title</th><th>Elements of Automobile Technology-I</th></tr>
<tr><td>2</td><td>Course Code</td><td>AM - 101</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Assistant Auto Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>This course is designed to understand the concept of automobile. Knowledge about engine working and role of various components of engine in performance of automobiles. Also this course enhance deep and vigorous understanding of fuel supply system, cooling system, lubrication system, transmission system and steering geometry of a vehicle and other major components of automobile.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand the concept of automobile and its major components.
CO2. Knowledge about working of engine and role of various components of engine in performance of automobiles.
CO3. Enhance understanding of fuel supply system, cooling system, lubrication system, transmission system and steering geometry of a vehicle.
CO4. Recognize components of automobile
CO5. Knowledge about basics of automobile air conditioner and emission system.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

Introduction to automobile, Classification of automobiles, Major components of automobile – Basic structure, power plant, transmission system, auxiliaries and controls.

UNIT– II

Introduction to Engines, Engine terminology, Basic engine parts and their uses

UNIT– III

Introduction to fuel system of petrol and diesel engines, Cooling system, Lubrication, Clutch, Gearbox, Brakes and Suspension system

UNIT-IV

Introduction to chassis, Wheels and Tyres, Propeller shaft, Rear axle assembly, Steering and Front axle, Automobile Emission Control, Heating and Air Conditioning System

Practical

1. Study and identification of automobile components, engine parts and their uses.
2. Introduction and demonstration of clutch and gearbox.
3. Study working of brakes (Hydraulic and Mechanical) and suspension of automobiles.
4. Familiarization with fuel system of both petrol and diesel engine, simple carburetor, ignition system of petrol engine.
5. Understanding working of cooling system (Air cooling and Water cooling) and lubrication system in automobiles.
6. Study of automotive heating and air conditioning system

13 Suggested Resources for Reading

1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Automotive Electronics and Electric System</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM – 102</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Assistant Auto Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>This course is designed for comprehensive study of AC and DC circuits that are required in automobile electrical and electronic systems. Students will also learn about construction, working of transformer and principle of different indicating instruments. Knowledge about generation of alternating electromotive force, concept of EMF, root mean square and average value of AC current also included in this course.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand concept of AC and DC circuits that are required in automobile electrical and electronic systems.
CO2. Theoretical and practical exposure about construction and working of transformer.
CO3. Knowledge about different indicating instruments.
CO4. Understand concept of EMF in detail.
CO5. Practical exposure of multimeter to measure AC and DC voltage and currents in the circuit.
…….</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
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3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

Electricity, Elements of electricity, Ohm's Law, Electric circuits, DC measuring instruments, Action of electric current, Voltage generation, Electrochemical cells

UNIT– II

Introduction to magnetic and electric fields, Faraday's law of electromagnetic induction, Self and Mutual inductance, Coupling coefficient, Voltage regulation and its applications

UNIT– III

Basic principle and classification of indicating instruments, Digital and analogue multimeter, Measurement of power energy, Resistance control and protection device, Circuit breaker, Fuse, MCB

UNIT-IV

Concept of EMF, Induced EMF and its types, Concept of resonance in series LCR circuit, Principle, construction and working of transformer, Core type and shell type transformers

Practical

1. Use of multimeter to measure AC and DC voltage and currents in the circuit.
2. Study of voltage, current and resonance in different circuits using Ammeter, Voltmeter, Galvanometer.
3. Study of circuit breaker and MCB used in automobiles, transformer's working, construction and its applications
4. Verification of ohm's experimentally.
5. Study the function of CRO.

13 Suggested Resources for Reading

1. Fundamentals of electrical and Electronics Engineering by B.L. Theraja, S. Chand and company, New Delhi.
2. Basic Electronic and electrical engineering by Bhattacharya S.K., Pearson education
3. Principles of electrical engineering by Gupta B.R., S. Chand and company, New Delhi.

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Automotive Tools</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM – 103</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table>a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Assistant Auto Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>This course is designed to provide students with deep understanding of automotive tools. They will also learn about the working of various power operated tool. Knowledge about threads, fasteners, nut, bolts, bearings and belts. Concept of service information and measuring instruments are also included in this course.
.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand automotive tools and power operated instruments.
CO2. Deep knowledge about threads, fasteners, nut, bolts, bearings and belts.
CO3. Practical exposure to operation of power operated tools.
CO4. Understand service information and measuring instruments.
CO5. Practical experience to use vernier caliper and screw gauge.
…….</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

Introduction to automotive tools, Hand tools, Wrenches, Screw driver, Pliers, Hammers and Mallets, Punches, Metal working tools, Hacksaws, Drills, Taps and Dies, Power tools – Electric, Hydraulic and Compressed air, Bench vise

UNIT– II

Fasteners, Threaded Fasteners, Thread size and designations, Washers, Torqueing Bolt, Removing of broken bolts, Gaskets, Adhesives, Sealers, Seals, Hoses and Lines, Belts and Bearings

UNIT– III

Introduction to service information, Steps in engine services, Safety tips for using hand tools, Safety tips for technicians, Engine service tools and equipment

UNIT-IV

Measuring instruments, Meters, testers and analysers, Testing safety rules, Using welding equipment, Shop safety, Fire prevention, Safety in hoisting a vehicle, Disposing of automobile wastes

Practical

1. Study and identification automotive tools
2. Introduction and demonstration of Fasteners, Washers, Torqueing Bolt, Gaskets, Adhesives, Sealers, Seals, Hoses and Lines, Belts and Bearings
3. Study operation of power operated tools
4. Calibration of vernier caliper, height/depth guage and screw guage.
5. To measure length, breath, thickness, depth, height, with vernier calipers

13 Suggested Resources for Reading

1. Automobile Engineering III by P.S. Gill Published by S.K. Kataria and Sons, New Delhi
2. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
3. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi

Note: Latest edition of text books may be used.

Semester II

<table>
<tr><td>1</td><td>Course Title</td><td>Elements of Automobile Technology-II</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM – 201</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table>a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Assistant Auto Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>The students will understand how engines are classified either on the basis of valve arrangement, number of strokes, cylinder arrangement, fuel used, engine cycles etc. Students should be able to understand about the working of petrol and diesel engines along with their merits and demerits. They will also gain knowledge of various components of engine and also how they affect the performance of an engine.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Knowledge about engine classification in detail.
CO2.Understand working of petrol and diesel engine along with its merits and demerits.
CO3.Practical exposure of major components used in engines.
CO4. Knowledge about cylinder block, cylinder head, piston, piston rings in detail.
CO5.Deep understanding of engine performance affected by size and material of engine components.
…….</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

Classification and types of engines based on engine cycles, Number of strokes. Fuel use, Number and Arrangement of cylinder, Valve arrangement, Type of valve. Special type engine- Square engine, Wankel engine, Atkinson cycle engine, Miller cycle engine and Electrical engine

UNIT– II

Petrol engine:- Introduction, merits and demerits. Diesel engine:- Introduction, merits and demerits. Valve timing of two stroke and four stroke engine

UNIT– III

Engine construction I:- Cylinder blocks, Cylinder head, Gasket, Piston, Piston ring, Cylinder liner, Manifold, Piston pumps, Connecting rods

UNIT-IV

Engine construction II:- Vibrant on damper, Camshaft, Engine bearings, Timing belt/chain, Balanced weight, Flywheel, Crankcase, Crankshaft and Valve

Practical

1. Introduction and demonstration of engine cycles, number of strokes, number of cylinder and its arrangements.
2. Familiarization of petrol and diesel engines used in automobiles
3. Study and demonstration of engine construction and components i.e piston, piston ring, cylinder block, cylinder head, cam shaft, crank shaft, vibrant damper manifold and timing chain/belt.
4. Study of construction and working principles of Three phase induction motor, Synchronous machines, DC machines

13 Suggested Resources for Reading

1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi

Note: Latest edition of text books may be used.

1	**Course Title**	**Advanced Electronic and Auto Electric System**
2	**Course Code**	**AM – 202**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical's** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Assistant Auto Technician
8	**Course Objectives**	The objective of this course is to make students understand about construction and working principle of DC machines, three phase induction motor and synchronous motor. This will also enable the students to understand about diodes, number system, microprocessors, flip flops, amplifiers, oscillators and multi vibrators. Knowledge about major electrical components used in automobiles.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** Knowledge about working of DC machines, three phase induction motor and synchronous motor**.** **CO2.**Practical exposure to study power and energy measurement using energy meter**.** **CO3.**Understand working of microprocessors, operational amplifiers, oscillators, multivibrators**.** **CO4.** Exposure to study characteristics of BJT and working of LCR circuit practically. **CO5.** Understand working of engine starter, generator/alternator, ignition coil and other electrical components**.** **…….**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

UNIT– I

DC machines (motors and generators), Three phase induction motor, Synchronous machines (motors and generators): construction and working principles

UNIT– II

Principle of operation, characteristics and applications of PN junction diode, Rectifiers, Zener diodes, bipolar junction transistors, Principle of operation, characteristics and applications of FET

UNIT– III

Binary, Octal and hexadecimal number system and its arithmetic operations, Introduction to microprocessors, operational amplifiers, oscillators, multivibrators, Testing ICs, Micro phone and loud speakers, LEDs, LCDs .

UNIT-IV

Introduction to engine starter, generator/alternator, ignition coil and other electrical components (fuses, relays, lighting system) used in automobiles

Practical

1. Study characteristics of PN junction diode.
2. Study of Full and Half wave Rectifiers.
3. Study of V-I characteristics of Zener diodes.
4. Characteristics of BJT and working of LCR circuit.
5. Study power and energy measurement using energy meter.

13 Suggested Resources for Reading

1. Fundamentals of electrical and Electronics Engineering by B.L. Theraja, S. Chand and company, New Delhi.
2. Basic Electronic and electrical engineering by Bhattacharya S.K., Pearson education
3. Principles of electrical engineering by Gupta B.R., S. Chand and company, New Delhi.

Note: Latest edition of text books may be used.

1	**Course Title**	**Automobile Physics**
2	**Course Code**	**AM – 203**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + … hours of practical'**s (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** / **Internal Theory: 20** a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Assistant Auto Technician
8	**Course Objectives**	The course under automobile physics has been designed to give the basic knowledge about the displacement, velocity and acceleration of automobiles. They also learn about kinetic energy potential energy and fraction associated with the automobiles. Fundamentals of AC , DC , resistance , conductance and gates (AND OR, NOT) are also included in this course. .
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** Understand layout of electrical system in Automotive**.** **CO2.** Knowledge about the displacement, velocity and acceleration. **CO3.** Exposure to binary number system**.** **CO4.** Knowledge about colour code for carbon resistors **CO5.** Deep understanding of capacitors (Parallel and Series). **…….**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

Difference between Alternating current and Direct current, Layout of electrical system in Automotive, Automotive wiring and its forms, Kirchoff's Law

UNIT– II

Newton's Law of Motion, Distance, Displacement, Velocity, Speed, Acceleration, Variable force, Kinetic and potential energy, Work energy theorem, Static and kinetic friction, Rolling friction, Lubrication

UNIT– III

Charge and its properties, electric field, potential, capacitor, parallel plate capacitors, capacitance, combination of capacitor in series and parallel, Resistance, Electrical resistivity, Conductivity, Resistor in series and parallel combination, Colour code for Carbon resistors

UNIT-IV

Binary number system, Addition and subtraction for binary numbers, OR gate, AND gate, NOR gate, NAND gate and XOR gate and their truth tables

13 Suggested Resources for Reading

1. Modern abc of Physics +1 and +2 by A. Gupta, Modern Pub. Jalandhar.
2. JBD Physics +1 and +2 by S.K. Gupta, K.L. Gosian and B.S. Satyal

Note: Latest edition of text books may be used.

Semester III

<table>
<tr><td>1</td><td>Course Title</td><td>Thermodynamical Physics</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-301</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>60 hours of lectures for Theory + … hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th></tr><tr><td>80</td><td>20</td></tr></table>a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed for comprehensive study of Thermodynamics’ involved in automobile engines. Students will also learn about basic laws of thermodynamics. In depth knowledge about Carnot Cycle (Reversible and Irreversible), Steam Engine, Heat Engine and Refrigeration.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand the concept of Thermodynamic system.
CO2. Knowledge about working of engines and role of thermodynamics in performance of automobiles.
CO3. Understanding various Laws of Thermodynamics involved in automobile engines.
CO4. Understand the concept Carnot Heat Engine.
CO5. Knowledge about basics processes in Thermodynamics.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT – I Thermodynamic system, Zeroth law of thermodynamics, Concept of heat, quasi static process, Heat and work- a path function, Comparison of heat and work, First law of thermodynamics and its applications, Isothermal, Adiabatic, Isochoric and Isobaric process. UNIT – II Gas equation during adiabatic process, Work done during adiabatic and isothermal elasticity's, Reversible and irreversible process, Second law of thermodynamics. UNIT – III Carnot's reversible engine and refrigerator, Carnot's theorem, Thermodynamic scale of temperature, Absolute zero on work scale, Ideal gas scale, Entropy and second law of thermodynamics. UNIT – IV Steam engine, Otto cycle, Internal combustion engine, Diesel engine, Multi cylinder engines, Third law of thermodynamics, Change in entropy in reversible process (Carnot's cycle)
12	**Suggested Resources for Reading**	1. Modern abc of Physics +1 and +2 by A. Gupta, Modern Pub. Jalandhar. 2. JBD Physics +1 and +2 by S.K. Gupta, K.L. Gosian and B.S. Satyal **Note: Latest edition of text books may be used.**

1	**Course Title**	**Automobile Technology-I**
2	**Course Code**	AM-302
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours** of **lectures** for **Theory + 6 0 hours of practical's** (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7	**Course Objectives**	This course is designed to understand the concept of Clutch and Gear Box. Knowledge about clutch actuating mechanism and role of various components of clutch in performance of transmission system. Also this course enhances deep and vigorous understanding of gear box, automatic transmission and transmission troubleshooting.
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the concept of clutch and its major components. CO2. Knowledge about working of clutch actuating mechanism. CO3. Enhance understanding of gear box and its types. CO4. Recognize components of automatic transmission. CO5. Knowledge about troubleshooting in transmission system.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

11	**Course Content**	UNIT – I Introduction to Clutch, Principle, functions and requirements of clutch, Main parts of clutch, Types of clutch. UNIT – II Clutch actuating mechanism, Clutch constructor, Function of clutch disc, Clutch lining material, Clutch troubleshooting. UNIT – III Function of transmission system, Necessity of transmission, Types of transmission, Gear selector mechanism, Four-wheel drive, Transmission troubleshooting. UNIT – IV Automatic transmission, Epicyclic gearbox, Torque converter, AMT, CVT, DCT/DSG, Gear reduction, Overdrive, Introduction to various speed gearboxes, Comparison of transmission. **Practical work:** 1. Practice on replacing clutch plate and other major components. 2. Practice on disassembling, assembling and fault diagnosing of gear box. 3. Study of gear shift mechanism, and replacement procedure of gear shift mechanism. 4. Dismantle, assemble and service differential and transfer case. 5. Demonstration of automatic transmission.
12	**Suggested Resources for Reading**	1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, new delhi **Note: Latest edition of text books may be used.**

<table>
<tr><td>1</td><td>Course Title</td><td>Fundamentals of Petrol Engine</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-303</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical's
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed to provide students with deep understanding of petrol engines. They will also learn about the working of ignition system. Knowledge about fuel supply system and its components. Concept of fuel injection system and CNG system is also covered in this course.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand combustion process in petrol engines and petrol fuel requirements.
CO2. Deep knowledge about working of petrol engines.
CO3. Practical exposure to operation of ignition system.
CO4. Understand fuel supply system and its components.
CO5. Knowledge about fuel injection and CNG system.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT – I Introduction to petrol engine, Properties of petrol, Octane number and rating, Combustion process in petrol engine, Types of combustion Chambers. UNIT – II Ignition system – function and requirement of ignition system, Types of ignition system, Ignition timing, Spark plug – construction, requirements and air gap setting, Electronic ignition system, Programmed ignition system. UNIT – III Methods of fuel supply system, Components of fuel supply system, Carburation, Functions of carburetor, MPFI system, Throttle body injection, Sequential fuel injection and Gasoline direct injection. UNIT – IV Throttle valve, Mixture control devices, Air cleaner, Types of inlet manifolds, Operation of CNG vehicles, Components of CNG kit, Troubleshooting in petrol engine **Practical work:** 1. Familiarization to working and combustion process of petrol engine and types of combustion chambers. 2. Study of ignition system and its components 3. Study methods of fuel supply system and its major components 4. Practice on operation of carburetor and MPFI system 5. Study and practice on operation of throttle body
12	**Suggested Resources for Reading**	1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, new delhi **Note: Latest edition of text books may be used.**

Semester IV

1	**Course Title**	**Lubrication System, Supercharging and Electric vehicles**
2	**Course Code**	AM-401
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours** of **lectures** for **Theory + … hours of practical's** (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** / **Internal Theory: 20** a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Course Objectives**	The students will understand about working of cooling system used in automobiles. Students should be able to understand about lubricants and types of lubrication system. They will also gain knowledge of supercharger and turbocharger. Also knowledge about electric vehicles.
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1.Knowledge about working of cooling system used in automobiles. CO2.Exposure to lubricants and types of lubrication system. CO3. Understand working of supercharger and turbocharger. CO4. Exposure to working of electric vehicles. CO5. Knowledge about batteries and motors used in electric vehicles.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

11	**Course Content**	UNIT – I Function of cooling system, Optimal temperature requirement of an engine, Types of cooling system- Air cooling and water cooling, Need for cooling system, Parts of cooling system - radiator, cooling fan, pressure cap, thermostat, pump, Water jackets, Cooling water additives, Replacement procedure and interval for coolant. UNIT – II Function and classification of lubricants, Viscosity of lubricant, properties of lubricant, Additives used in lubricants, Replacement procedure and interval for various lubricants. Introduction to lubrication system of engine, System of lubrication – petroil system, pressure system, dry sump system, Oil pump and types of oil pump, Crank case ventilation, Oil cooler, Oil filter. Importance of service manual in replacement of lubricant. UNIT – III Purpose and effect of supercharging, Merits and demerits of supercharging, Components of supercharger, Turbocharging, Comparison between turbo charging with supercharging, Turbo lag, Method of turbocharging and components of turbocharger – turbine and compressor, Limitations of turbocharging, Cooling of turbocharger and supercharger. Basic requirements for better performance of turbo for longer period. Trouble shooting in turbocharger. UNIT – IV Electric vehicles – components of electric vehicle, Types of motors used in electric vehicles, Energy storage batteries and there types, AC and DC charging, Normal charging , Fast charging, Wireless charging, Regenerative braking system.
12	**Suggested Resources for Reading**	1. Automobile Engineering III by P.S. Gill Published by S.K. Kataria and Sons, New Delhi 2. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 3. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, new delhi **Note: Latest edition of text books may be used.**

1	**Course Title**	**Automobile Technology-II**
2	**Course Code**	AM-402
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours** of **lectures** for **Theory + … hours of practical's** (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** / **Internal Theory: 20** a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Course Objectives**	This course is designed to understand the concept of Steering system and Drive line. Knowledge about steering geometry and role of various components of steering system in performance of control system. Also this course enhances deep and vigorous understanding of differential.
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the concept of steering geometry. CO2. Knowledge about working of steering system. CO3. Enhance understanding of power steering system CO4. Recognize components of drive line. CO5. Knowledge about working of differential and troubleshooting in differential.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

11	**Course Content**	UNIT – I Functions of the steering system, Steering gears, Steering mechanism used in some Indian vehicles, Steering linkage, Steering Wheel and column, Front axle UNIT – II Steering Heads, Steering geometry, Wheel alignment, Adjusting steering angles, Power steering, Understeering and Oversteering, Steering lock, Turning radius, Steering troubleshooting UNIT – III Introduction to real axle assembly, Types of rear axle assembly Purpose of the final drive, Final drive requirements, Transfer box, Torque tube drive, Axle housing, Maintenance of real axle UNIT – IV Transaxles- Manual and Automatic, Differential- Requirement and working, Troubleshooting in differential, Differential lock, Mechanical locking differential, Brake locking differential
12	**Suggested Resources for Reading**	1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi **Note: Latest edition of text books may be used.**

<table>
<tr><td>1</td><td>Course Title</td><td>Fundamentals of Diesel Engine</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-403</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical's
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed to provide students with deep understanding of diesel engines. They will also learn about the working of fuel supply system and its components. Knowledge about fuel injection system. Concept of automobile emissions and major constituents of automobile emissions is also covered in this course.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand combustion process in diesel engines and diesel fuel requirements.
CO2. Deep knowledge about working of diesel engines.
CO3. Practical exposure to operation of fuel injection system.
CO4. Understand fuel supply system and its components.
CO5. Knowledge about automobile emissions.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT – I Introduction to combustion process in diesel engines, Properties of diesel fuel, Combustion process in CI engine, Diesel knock. UNIT – II Types of CI combustion Chambers, Open combustion chambers, Free combustion chamber, Turbulence combustion chamber, Air cell combustion chamber, Energy cell combustion chamber. UNIT – III Introduction to fuel injection for diesel engine, Fuel injection system, Computer controlled diesel injection, Fuel injection pump for diesel engine, Diesel injector, Governor. UNIT – IV Automobile Emission, Constituents of exhaust, Sources of pollutants in an automobile, Control approaches for automobile emissions, Alternative automobiles fuels. **Practical work:** 1. Identification and study of diesel engine parts, diesel injector, diesel fuel pump and common rail. 2. Perform testing of fuel injector and fuel pump 3. Study and perform reconditioning of diesel engine 4. Practice on cleaning Exhaust Gas Recirculation system 5. Practice on cleaning catalytic converter
12	**Suggested Resources for Reading**	1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi. 3. Automobile Engineering I by P.S. Gill Published by S.K. Kataria and Sons, New Delhi **Note: Latest edition of text books may be used.**

1	**Course Title**	**Workshop 1**
2	**Course Code**	AM-404
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours** of **lectures** for **Theory + 120 hours of practical's** (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Practical: 80** / **Internal Practical: 20** a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
7	**Course Objectives**	The objective of this course is to make students practically understand about steering geometry and components of steering system. This will also enable the students to understand about how steering system maintenance procedure is done. Practical exposure to replacement procedure of various lubricants and oil filters.
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Practical understanding of steering geometry and components of steering system. CO2. Deep knowledge about setting wheel alignment and adjusting steering angles. CO3. Practical exposure to fault diagnosis process in cooling system and lubrication system. CO4. Understand fault diagnosis procedure in turbo charger and servicing of intercooler. CO5. Knowledge about power unit used in power steering system.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	The syllabus has been divided into eleven experiments. For practical exam there should be at least two experiments for practical performance of 60 marks including Viva – Voce and one written part of 20 marks.

11	**Course Content**	**Practical work:** 1. Study of steering system, steering gears, steering mechanism used in some Indian vehicles. 2. Study of steering linkage, steering wheel and column and front axle 3. Perform on setting wheel alignment and adjusting steering angles. 4. Study power steering system (hydraulic and electronic) and steering lock 5. Practice on repairing steering rack assembly. 6. Reassemble all components of cooling system and fault diagnosing for over heat and under heating of engine. 7. Perform cooling system service (thermostat, radiator, cooling fan and pump). 8. Practice on cooling system pressure test , testing of cooling fan and clutch, heater core and coolant temperature warning light. 9. Practice on replacing different lubricants and filters of automobile. 10. Study and practice on oil pressure problems, oil pump service and oil pressure indicator. 11. Practice on fault diagnosis in turbo charger and servicing of intercooler.
12	**Suggested Resources for Reading**	1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi. 3. Automobile Engineering II by P.S. Gill Published by S.K. Kataria and Sons, New Delhi **Note: Latest edition of text books may be used.**

Semester V

<table>
<tr><td>1</td><td>Course Title</td><td>Solar Cells in Automobile Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM – 501</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>60 hours of lectures for Theory + …..hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th></tr><tr><td>80</td><td>20</td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed to understand the concept of Solar Cells. Knowledge about solar energy and its role in automobiles. Also this course enhance deep and vigorous understanding of components and design of solar cells along with other performance related parameters in automobiles.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand the concept solar energy.
CO2. Knowledge about working of solar cells.
CO3. Enhance understanding of components and design of solar cells in automobiles
CO4. Recognize Solar cell vehicle layout
CO5. Knowledge about basics performance related parameters.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11 Course Content

UNIT– I

Introduction to solar energy, nature of solar energy, solar energy and environment, method of using solar energy, future scope of solar energy

UNIT– II

Introduction to solar cell, principle, construction and working, applications of solar cells, performance of solar cell, components and design of solar cells in automobiles

UNIT– III

Module fabrication, module interconnection, basic of load estimation, merits and demerits of solar energy in automobile industry

UNIT-IV

Solar cell vehicle layout, performance of vehicle, transmission requirements, vehicle performance, energy consumption, advantage and limitations of solar control system, safety and challenges of hybrid vehicles

12 Suggested Resources for Reading

1. Solar cell Technology and Applications by A.R. Jha, CRC Press, London
2. Solar Cells: Materials, Manufacture and Operation by Augustin McEvoy, L. Castaner, Tom Markvart, Elsevier, Paris
3. Physics of solar cells 3rd Edition by Peter Würfel, Wiely, London

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Automobile Technology-III</th></tr>
<tr><td>2</td><td>Course Code</td><td>AM 502</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>60 hours of lectures for Theory + …. hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th></tr><tr><td>80</td><td>20</td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed for comprehensive study of automobile chassis and denting/painting in automobiles. Students will also learn about role of NVH. Knowledge about paints, primers, sand papers and spray paint gun also included in this course.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand concept of chassis layout and frames.
CO2. Theoretical exposure about types/function of body and panels.
CO3. Knowledge about paints, primers, sand papers and spray paint gun
CO4. Understand concept of denting and painting.
CO5. Exposure about Automobile Noise Vibration and Harshness</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11 Course Content

UNIT– I

Chassis Layout and Frames: - Introduction, classification of chassis, types of frames and sub frames, Automotive materials, Types of automobile wheels – steel wheel wire wheel, alloy wheel, materials for wheels, Types of tyres and carcass, tyre head, tyre selection, tyre service parameters, tyre maintenance, materials used in tyres

UNIT– II

Introduction on types, function of body and panels, Procedure for inspection, removing and refitting body components, Safety considerations during denting and painting. Procedure of denting, rust treatment, paint production.

UNIT– III

Procedure for applying body filler, putty, base coat primer and final paint, Types of automotive paints, colour codes primers, tools for denting painting and sand papers, Spray painting gun and drive, Requirements of body care and maintenance.

UNIT-IV

Automobile NVH- Role of noise vibration and harshness in automotive design and development. Sources of vibration and noise in automobile. Methods to control NVH level.

12 Suggested Resources for Reading

1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi
3. Automobile Engineering II by P.S. Gill Published by S.K. Kataria and Sons, New Delhi

Note: Latest edition of text books may be used.

1	**Course Title**	**Workshop – 2**
2	**Course Code**	**AM – 503**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**…. hours of lectures for Theory + 120 hours of practical's** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Practical: 80** / **Internal Practical: 20** a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. **c.** Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
7	**Course Objectives**	The objective of this course is to make students practically understand about denting and painting procedures. This will also enable the students to understand about how cooling system and lubrication system maintenance is done. Practical exposure on fault diagnosis in turbo charger
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** Understand automotive denting and painting procedures**.** **CO2.** Deep knowledge about paints, primers, sand papers and spray gun**.** **CO3.** Practical exposure to maintenance of cooling system**.** **CO4.** Understand lubrication system and lubricants. **CO5.** Practical experience to fault diagnosis and maintenance of turbocharger**.** **…….**
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	The syllabus has been divided into eleven experiments. For practical exam there should be at least two experiments for practical performance of 60 marks including Viva – Voce and one written part of 20 marks.

11 Course Content

Practical

1. Dismantle and assemble various parts of body shell and windscreen.
2. Practice on operating air compressor, paint gun, grinder and rubbing/polishing machine.
3. Practice on operation of denting tools
4. Reassemble all components of cooling system and fault diagnosing for over heat and under heating of engine.
5. Perform cooling system service (thermostat, radiator, cooling fan and pump).
6. Practice on cooling system pressure test, testing of cooling fan and clutch, heater core and coolant temperature warning light.
7. Study lubrication system service and troubles.
8. Practice on replacing different lubricants and filters of automobile.
9. Study and practice on oil pressure problems, oil pump service and oil pressure indicator
10. Dismantle and assemble turbo charger.
11. Practice on fault diagnosis in turbo charger and servicing of intercooler.

13 Suggested Resources for Reading

1. Auto Body Repair Technology, James E. Duffy, Cengage Learning India Private Limited, Delhi, India
2. Automobile Engineering by P.S. Gill Published by S.K. Kataria and Sons, New Delhi
3. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
4. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi

Note: Latest edition of text books may be used.

<table>
<tr><th>1</th><th>Course Title</th><th>Modern Electric and Hybrid Vehicles I</th></tr>
<tr><td>2</td><td>Course Code</td><td>AM – 504</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The students will understand how electric and hybrid vehicles works. Students should be able to understand about drive train topologies in EV’s and HEV’s. They will also gain knowledge of electric drive components used in electric and hybrid vehicles.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Knowledge about electric and hybrid electric vehicles in detail.
CO2.Understand Power flow control in electric and hybrid electric drive train topologies
CO3.Exposure of electric drive components used in electric and hybrid vehicles
CO4. Knowledge about Regenerative Braking System.
CO5.Deep understanding of vibration energy harvesters.
…….</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11 Course Content

UNIT– I

Introduction to electric and hybrid electric vehicles, History of hybrid and electric vehicles, Electrical basics, Motor and generator basics, Social and environmental importance of electric and hybrid electric vehicles

UNIT– II

Basic concept of electric and hybrid traction, Introduction to various electric and hybrid electric drive train topologies, Advantages and disadvantages, Power flow control in electric and hybrid electric drive train topologies

UNIT– III

Introduction to electric drive components used in electric and hybrid vehicles, Electric motor requirements, Direct Current (DC) motors (Brushed and Brushless), Power converters, Drive controllers.

UNIT-IV

Introduction and need of Regenerative Braking System, Advantages and disadvantages of RBS, Working of RBS, Concept of Regenerative Braking using Piezoelectric material, Using shock absorbers as vibration energy harvesters.

Practical's

1. Understand working of different configurations of electric vehicles.
2. Understand hybrid vehicle configuration and its components.
3. Understand the properties of batteries and its types.
4. Understand of electric vehicle drive systems.

12 Suggested Resources for Reading

1. Electric & Hybrid Vehicles, A.K. Babu, Khanna Publishing House
2. Automotive Fuel Technology-Electric, Hybrid and Fuel-Cell Vehicles: Jack Erjavec & Jeff Arias
3. Electric and Hybrid Vehicles: Design Fundamentals: Iqbal Husain
4. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory and Design: Mehrdadehsani, Yimingao, Ali Emadi

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Off Road Vehicles and Automotive Fuels</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM - 505</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table>a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The objective of this course is to make students understand about farm power machinery and earth moving machinery. This will also enable the students to understand about working features of various types of Earthmoving machines. Knowledge about automotive fuels.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Knowledge about tractor’s along with its functions and applications.
CO2. Exposure to earth moving machinery.
CO3.Understand working features of various types of Earthmoving machines.
CO4. Exposure to of automotive fuels and scenario of conventional auto fuels.
CO5. Understand properties, production, storage, handling, performance and safety aspects in automotive fuels.
…….</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11 Course Content

UNIT– I

Introduction and classification of various off road vehicles, TRACTORS: General description; Specifications; functions and applications, Types and use of tractors and its equipment's

UNIT– II

Earth moving machinery, types of earth moving machinery and equipment's, Constructional details and working features of various types of Earthmoving machines, Combat Vehicles, Scrappers, graders, shovels and ditchers

UNIT– III

Classification of automotive fuels, Scenario of conventional auto fuels, Oil reserves of the world fuel quality aspects related to emissions, Technological up gradations required, Need for alternate fuel, Business driving factors for alternative fuels, Roadmap for alternative fuels, Alternate fuel development worldwide

UNIT-IV

Automotive Fuels: Properties, production, storage, handling, performance and safety aspects, advantages and disadvantages

Practical's

1. Study working of farm tractor
2. Perform servicing of engine and transmission of farm tractor
3. Study and practice on servicing of farm tractor implements
4. Study electrical components of farm tractor

12 Suggested Resources for Reading

1. Off-Road Vehicle Engineering Principles,Georing, C.E., Stone, D.W. Smith, P,K.Turnquist , ASAE2005

2. Construction, planing, equipment and methods, Robert L. Peurifoy, Tata McGraw Hil Publishing company Ltd.

3. Alternative Fuels, Thipse S.S., Jaico Publications.

4. Engine Emissions: Pollutant Formation and Advances in Control Technology Pundir B.P., Narosa Publications

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Advanced Automotive Fault Diagnosis - I</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM 506</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The course under Advanced Automotive Fault Diagnosis has been designed to give the basic knowledge about advanced diagnosis techniques. They also learn about various tools used in diagnosis process. Fundamentals of sensors and on board diagnostics are also included in this course.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand diagnosis process in Automotive.
CO2. Knowledge about mechanical and electrical diagnostic techniques.
CO3. Exposure to tools and equipment’s in diagnosis.
CO4. Knowledge about sensors and actuators.
CO5. Deep understanding of on board diagnostics.
…….</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11 Course Content

UNIT– I

Introduction to diagnosis, Safe working practices, Terminology, Diagnostic process, Diagnostics on paper, Real world diagnostics

UNIT– II

Mechanical diagnostic techniques, Electrical diagnostic techniques, Tools and equipment – Diagnostic scanner, Multimeters, Oscilloscopes, Emission testing

UNIT– III

Sensors – Introduction and diagnostics, Actuators, Testing of actuators, Engine waveforms, Communication networks – CAN, LIN, Flex ray, Time triggered CAN, Ethernet

UNIT-IV

Introduction to on board diagnostics, Petrol/Gasoline on board diagnostics, Misfire detection, Driving cycles, Future developments in diagnostic systems

Practical's

1. Study and perform diagnostics on paper and real world diagnostics
2. Study Mechanical and Electrical diagnostic techniques
3. Perform on fault diagnosis in communication networks
4. Perform on Misfire detection

12 Suggested Resources for Reading

1. Advanced Automotive Fault Diagnosis, Tom Denton, Fifth Edition, Abindon Oxon, New York
2. Automobile Engineering III by P.S. Gill Published by S.K. Kataria and Sons, New Delhi
3. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.

Note: Latest edition of text books may be used.

Semester VI

<table>
<tr><td>1</td><td>Course Title</td><td>Modern Vehicle Technology</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-601</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>60 hours of lectures for Theory + … hours of practical’s
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th></tr><tr><td>80</td><td>20</td></tr></table>a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed for comprehensive study of Modern Vehicles. Students will also learn about latest power plant technologies. In depth knowledge about Computer controlled vehicle management system.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand the concept of Modern Technologies in automobiles.
CO2. Knowledge about techniques used in reduction of pollution.
CO3. Understanding latest power plant technologies.
CO4. Understand Computer controlled vehicle management system.
CO5. Knowledge about various sensors.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT – I Automated vehicles, automatic parking, auto pilot mode, driving modes, electronic stability program, traction control system, emergency brake assist, electronic brake force distribution, 3d camera, antilock braking system, active suspension system, driver monitoring system, head up display, computer based automatic transmission, dynamic steering, communication between vehicles (smart cars), satellite control of vehicle operation for safe and fast travel, passenger comfort – recliner seats, ventilated seats and massaging seats. Computerized vehicle operation and control for better fuel economy and operation of vehicle like optimum speed and direction. UNIT – II Reduction of pollution - Internal and external pollution control through alternate fuels. Power plants- Refined fuel supply system, on board diagnosis for pollution control, catalytic converters, SCR technology, DPF and other filters for particulate emission. Bharat stage norms. UNIT – III Latest power plant technologies – Cam less engine, direct injection, active alternator, adaptive cruise control, cylinder deactivation, variable valve timing, continuously variable valve duration, hybrid vehicle, mild hybrid engines, hydrogen engine, advanced spark ignition, integrated inlet and exhaust manifold. UNIT – IV Computer controlled vehicle management system – ECM, EBCM, PCM and VCM. Sensors – Mass air flow sensor, engine speed sensor, coolant sensor, manifold absolute pressure sensor, camshaft position sensor, throttle position sensor, vehicle speed sensor, brake sensor, tyre pressure sensor, knock sensor, oxygen sensor.
12	**Suggested Resources for Reading**	1. Automobile Engineering by P.S. Gill Published by S.K. Kataria and Sons, New Delhi 2. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 3. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, new delhi **Note: Latest edition of text books may be used.**

1	**Course Title**	**Automobile Technology-IV**
2	**Course Code**	AM-602
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours** of **lectures** for **Theory + ... hours of practical's** (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** / **Internal Theory: 20** a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
7	**Course Objectives**	This course is designed to understand the concept of Braking. Knowledge about suspension system, its types and components. Also this course enhances deep and vigorous understanding of air conditioning and safety in automobiles.
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Understand the concept braking. CO2. Knowledge about suspension system and its components. CO3. Enhance understanding of safety in automobiles. CO4. Recognize components of air conditioning system in automobiles. CO5. Knowledge about new car assessment programs.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

11 Course Content

UNIT – I

Terms of braking – stopping distance and time, weight transfer during braking, braking efficiency, leading/trailing shoe of brakes, functions of a brake, requirements of a good braking system, Classification of brakes - drum brakes and disc brakes, Brake actuating mechanism- mechanical brake, hydraulic brake, pneumatic brake, vacuum brake and electric brake. Master cylinder, Tendom master cylinder, pressure regulator, wheel cylinder, brake fluid, brake fluid, trouble shooting in braking system.

UNIT – II

Function of suspension system, requirement for a good suspension system, sprung weight, un sprung weight, wheel rate types of suspension system – conventional and independent, springs- coil spring, rubber spring and leaf springs, shock absorbers, airmatic suspension, nitrox suspension, linkages, anti roll bar, torsion bar, maintenance of suspension system, inspection and service of suspension system, troubleshooting of suspension systems.

UNIT – III

Air conditioning – Introduction to refrigeration and vapour compression system. Effect of operating parameters in performance of A/C system. Types of refrigerants and their properties used in automobiles. Components of Air Conditioning, air management and heater system. Air distribution modes, ac zones – dual zone, four zone and temperature control system – automatic and manual.

UNIT – IV

Safety - Motor vehicle safety standards, active safety, passive safety, advanced materials and platforms for safety, Car crash test norms in Bharat NCAP, GNAP and ENCAP. Structural safety- Crumple zone, occupant compartment, side impact protection. Passenger Safety:- Seat Belts, air bags, other gadgets fixed in car for safe operations, safe driving, precaution in case of an accident, Some features for safety in modern cars – pre collision technology, driver alert, dynamic steering, automated emergency braking.

12 Suggested Resources for Reading

1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, new delhi
3. Automobile Engineering by Kipal S. Gill Published by S.K. Kataria and Sons, New Delhi

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Vehicle Performance and Garage Setup</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-603</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical's
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed to provide students with deep understanding of vehicle performance parameters. They will also learn about the vehicle testing techniques. Knowledge about garage and service stations. Understand concept of Motor vehicle act.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand vehicle performance parameters.
CO2. Deep knowledge about vehicle testing techniques.
CO3. Practical exposure to estimation of power requirement.
CO4. Understand process to setup Garage and Service stations.
CO5. Knowledge about Motor vehicle act.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT – I Vehicle performance parameters – handling, top speed, acceleration, fuel economy, comfort, braking, suspension system, ergonomics considerations, Drive train and power plant testing – engine performance, gear box and clutch working, Vehicle testing – track testing, crash testing and laboratory testing. Concept of bhp and torque in vehicles, methods of determining the engine horsepower and engine efficiency, Losses in engine. UNIT – II Vehicle testing – track testing, crash testing and laboratory testing. Concept of bhp and torque in vehicles, methods of determining the engine horsepower and engine efficiency, Losses in engine, Ratings used in automobile sector UNIT – III Introduction to Garage and Service stations, types of garages, location and layout of garage and service station, equipment for garage and service station, important services and equipment, standard service procedure for automobiles, record keeping and store management in garages. UNIT – IV Motor vehicle act – definition of vehicles, certifying and testing, transfer of ownership, vehicle registration. **Practical work:** 1. Estimation of power requirement for vehicle propulsion by taking actual vehicle example, On road fuel consumption test at different speeds 2. Practice on testing vehicle performance, transmission and engine efficiency. 3. Prepare layout of workshop according to category of workshop, inventory and equipment management in workshop stores 4. Prepare service procedure of automobiles and job card by inspecting all necessary things 5. Study requirements for insurance, transfer of ownership and vehicle registration
12	**Suggested Resources for Reading**	1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi 3. Automobile Engineering by Kipal S. Gill Published by S.K. Kataria and Sons, New Delhi **Note: Latest edition of text books may be used.**

1	**Course Title**	**Workshop 3**
2	**Course Code**	AM-604
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**… hours** of **lectures** for **Theory + 120 hours of practical's** (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Practical: 80** / **Internal Practical: 20** a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
7	**Course Objectives**	The objective of this course is to make students practically understand about fault diagnosis process in sensors. This will also enable the students to understand about to use scanning device. Practical exposure to servicing of automotive air conditioner and heating system. .
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1. Practical understanding of working on sensors. CO2. Deep knowledge about various valves used in automobiles and there fault diagnosis. CO3. Practical exposure to fault diagnosis process with scanning tool. CO4. Understand fault diagnosis procedure suspension system. CO5. Knowledge about servicing of automotive air conditioner and heating system.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	The syllabus has been divided into ten experiments. For practical exam there should be at least two experiments for practical performance of 60 marks including Viva – Voce and one written part of 20 marks.

11	**Course Content**	Practical's 1. Practice on fault diagnosis in sensors i.e cam position sensor, crank sensor, manifold absolute pressure sensor, knock sensor, air mass sensor,O_2 sensor, speed sensor, ABS sensor and motor. 2. Study working of EGR valve, solenoid valve and fuel metering valve. 3. Practice on fault diagnosis in valves. 4. Practice on using car scanner and diagnose trouble codes 5. Practice on operating pressure washer and vacuum cleaner used for cleaning and washing. 6. Practice on replacing of brake pads and brake shoes, caliper greasing, brake bleeding, changing brake fluid and checking of brake lines. 7. Assemble and disassemble shock absorber, wishbone, anti roll bar, linkages, replace rubber kit of suspension system. 8. Practice on inspecting vehicle including condition of engine, tyre life, paint condition and prepare inspection sheet for each part. 9. Practice on servicing of automotive air conditioner and heating system. 10. Study and demonstration of controls in refrigeration and perform leak testing.
12	**Suggested Resources for Reading**	1. Automobile Engineering III by P.S. Gill Published by S.K. Kataria and Sons, New Delhi 2. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 3. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, new delhi **Note: Latest edition of text books may be used.**

<table>
<tr><td>1</td><td>Course Title</td><td>Modern Electric and Hybrid Vehicles II</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-605</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical's
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed to understand the concept Hybrid Architecture and Power Plant Specifications. Knowledge about Sizing the Drive System and Energy Storage Technology. Also this course enhances deep and vigorous understanding batteries in EV's and HEV's.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand the concept of Hybrid Architecture.
CO2. Knowledge about Sizing the Drive System and Energy Storage Technology
CO3. Enhance understanding of Nonelectric Hybrid Systems
CO4. Recognize components of Electric and Hybrid vehicles.
CO5. Knowledge about Battery based energy storage</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT – I Hybrid Architecture and Power Plant Specifications: Series configuration locomotive drives- series parallel switching- load tracking architecture. Pre transmission parallel and combined configurations Mild hybrid- power assist- dual mode- power split power split with shift- Continuously Variable transmission (CVT)- wheel motors. Grade and cruise targets- launching and boosting- braking and energy recuperation- drive cycle implications. UNIT – II Sizing the Drive System and Energy Storage Technology: Matching electric drive and ICE; sizing the propulsion motor; sizing power electronics. Battery basics; lead acid battery; different types of batteries; battery parameters, Fuel cell vehicles UNIT – III Battery based energy storage: Battery basics, Lead acid (Pb-Acid) battery, Nickel-Cadmium (NiCd) battery, Nickel-Metal-Hydride (NiMH) battery, Lithium-ion (Li-ion) battery, Lithium-polymer (Li-poly) battery, Ultra capacitors. UNIT – IV Nonelectric Hybrid Systems: Short term storage systems flywheel accumulators. continuously variable transmissions hydraulic accumulators hydraulic pumps/motors pneumatic hybrid engine systems operation modes. **Practical's** 1. Understand of hybrid electric vehicles. 2. Understand Auxiliary systems including charging, starter motor, on board power supply, lighting and environmental sensing and conducting repairs. Repair & Replacement of Electric/ Hybrid Vehicle body 3. Repair & Replacement of Electric Vehicle Drive Train 4. Fault diagnosis & repair / replacement of Battery
12	**Suggested Resources for Reading**	1. Electric & Hybrid Vehicles, A.K. Babu, Khanna Publishing House 2. Automotive Fuel Technology-Electric, Hybrid and Fuel-Cell Vehicles: Jack Erjavec & Jeff Arias 3. Electric and Hybrid Vehicles: Design Fundamentals: Iqbal Husain 4. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory and Design: Mehrdadehsani, Yimingao, Ali Emadi **Note: Latest edition of text books may be used.**

<table>
<tr><td>1</td><td>Course Title</td><td>Automobile Electrical System</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-606</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical's
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is designed to provide students with deep understanding of electrical systems. They will also learn about the working of ignition system and its components. Knowledge about automobile fuses and relays. Knowledge of automobile lightning system.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Understand automobile wiring system
CO2. Deep knowledge about starter motor and generator(alternator).
CO3. Practical exposure to operation of various electrical components.
CO4. Understand ignition system and its components.
CO5. Knowledge about automobile lightning.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT – I Starting system - Principle, construction and working of starter motor. Series motor and its characteristics, Compound wound motor, Engine starting circuit, Starter switch - manual, solenoid, Factors affecting the starting of engines, Torque terms. Starting torque and power required, Motor efficiency, Armature reaction, Typical motor specifications UNIT – II Principle, construction and working of alternator and wiper motor, Alternators used in smart hybrid vehicles, Spark plugs-constructional details; types used in automobiles, conditions of working of spark plugs, Glow plugs of diesel engines UNIT – III Magneto-rotating armature and rotating magnet types, Electronic ignition of cars & motor-cycles (CDI), Idea of Distributor-less Direct ignition system, Types of fuses and relay used UNIT – IV Requirements of automobile lighting, wiring system of electrical components, Asymmetrical head light, dipper and full beam, care of headlamp, Lens cleaners. Dynamic headlight beam control, Advanced Front lighting system (AFS), Trouble shooting in electrical system. **Practical work:** 1. Dismantle study, assemble and check for proper working of engine starter, wiper motor, charging system. 2. Fault diagnosis in automotive wiring.. 3. Study of cranking circuit, battery life and maintenance, Practice on gravity testing, level of electrolyte and cleaning of battery terminals 4. Study and perform fault diagnosing in electronic ignition of cars 5. Dismantle, inspect and assemble the magneto of a 2-wheeler
12	**Suggested Resources for Reading**	1.Automotive Electricals and Electronics, A.K. Babu, Khanna Publishing House 2. Automotive Electrical Equipment: PL Kohli 3. Modern Electrical Equipment: AW Judge 4. Automotive Electrical Equipment: WH Crouse **Note: Latest edition of text books may be used.**

<table>
<tr><td>1</td><td>Course Title</td><td>Advanced Automotive Fault Diagnosis – II</td></tr>
<tr><td>2</td><td>Course Code</td><td>AM-607</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>30 hours of lectures for Theory + 60 hours of practical's
(1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table>a. The exam of this course will be of 3 hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The objective of this course is to make students understand about diagnostics. This will also enable the students to practically understand about procedures of fault recognizing in various components. Exposure to diagnosis in all electrical components.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Practical understanding of diagnosis in Engine, fuel supply system, ignition system and emission system.
CO2. Deep knowledge about fault detection ABS system
CO3. Practical exposure to fault diagnosis process related to immobilizer and car security system
CO4. Understand fault diagnosis Heating ventilation and air conditioning
CO5. Knowledge about modern diagnostic techniques.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>10</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
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3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

11	**Course Content**	UNIT - I Diagnostics – Engine, fuel supply system, ignition system, emission system, fuel injection system, diesel injection system, engine management, cooling system, lubrication system UNIT - II Diagnostics – Brakes, antilock brakes, traction control, steering and tyres, suspension, active suspension UNIT - III Diagnostics – Multiplexing, lighting, auxiliary, in car entertainment, security and communications, body electrical systems, instruments UNIT - IV Diagnostics –Heating ventilation and air conditioning, cruise control, air bags and belt tensioners, manual transmission, automatic gearbox **Practical's** 1. Diagnose fault in engine management system 2. Study and perform fault detection and rectifying various faults in ABS system 3. Diagnose faults related to immobilizer and car security system 4. Study and perform fault diagnosing in heating ventilation and air conditioning
12	**Suggested Resources for Reading**	1. Advanced Automotive Fault Diagnosis, Tom Denton, Fifth Edition, Abindon Oxon, New York 2. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar. 3. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi. 4. Automobile Engineering II by P.S. Gill Published by S.K. Kataria and Sons, New Delhi **Note: Latest edition of text books may be used.**